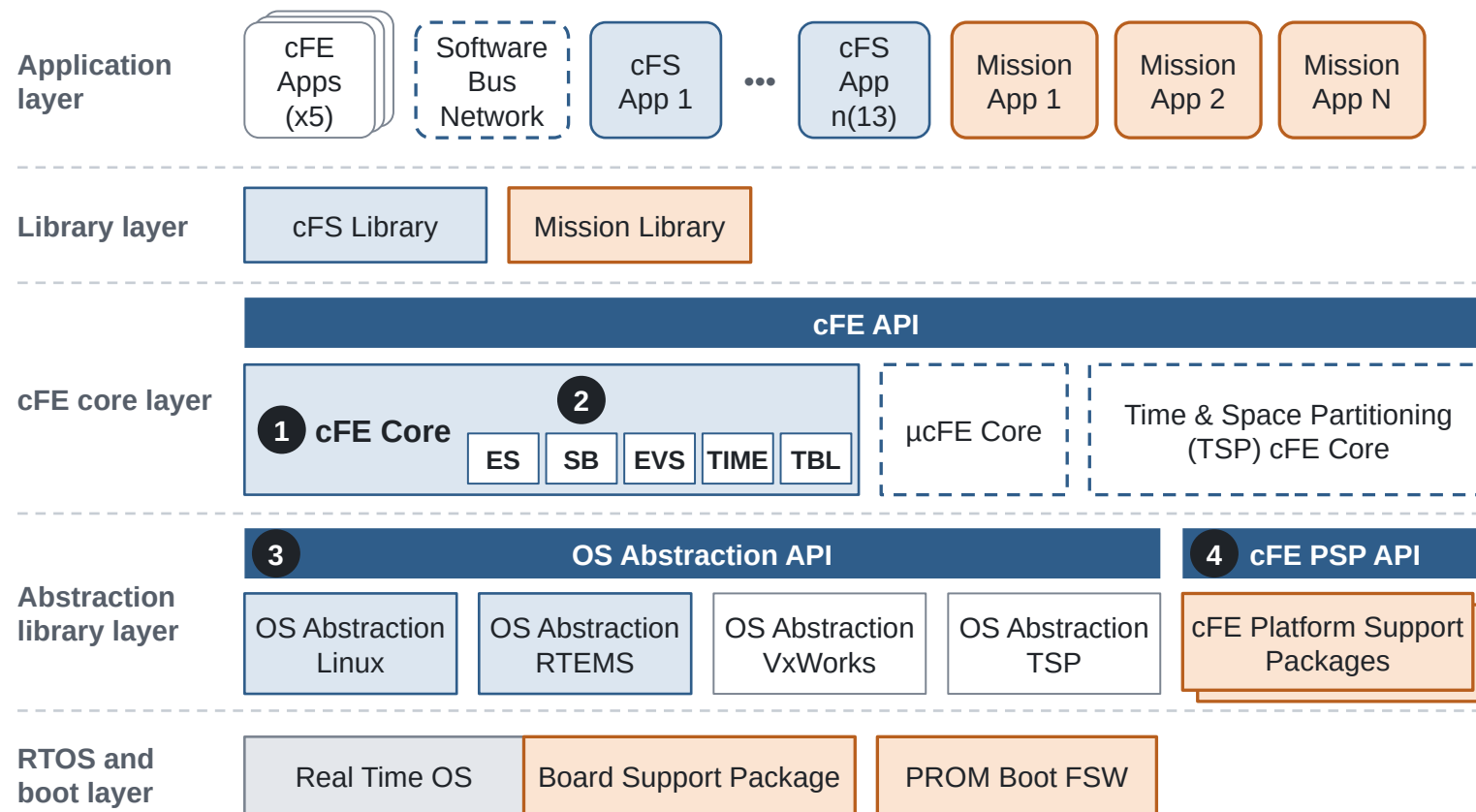


The OSAL and PSP hide the real-time OS and the board, so the cFE and the apps above it stay platform independent



- 1 cFE Core services** pp. 25, 45, paraphrase
Provide the services common to most flight apps: ES, SB, EVS, TIME and TBL
- 2 Software bus (SB)** p45
“Provides an application publish/subscribe messaging service”
- 3 OSAL (OS Abstraction API)** p23, paraphrase
One API to the cFE, whatever real-time OS runs underneath
- 4 PSP (cFE PSP API)** p24, paraphrase
One API to the avionics hardware and board support package

Status as of August 2015

- | | |
|------------------------------|-------------------|
| Open source, NASA maintained | Mission developed |
| In development | Third party |
| No status color in source | Layer API |

Source: diagram rebuilt from McComas et al., NASA Goddard, Aug. 2015, NTRS 20150018075, p8. Service acronyms and order: 2021 memo, printed p45 (the 2015 deck lists the same five services on p9). Roles: NASA/TM-20205000691/REV 1, Jan. 2021, NTRS 20205011588, printed page on each heading. Dark bars: each layer's standard API (2023 overview, NTRS 20230002444, slide 8).